

Experiment 4

Problem:

Find all prime numbers between two numbers.

Student Details:

Name: madhiya zaid javed Roll No: 36 UIN: 251M040 Division: F

Code (C):

```
#include <stdio.h>
#include <math.h>
int isPrime(int n) {
    if(n<=1) return 0;
    if(n==2) return 1;
    if(n%2==0) return 0;
    for(int i=3;i<=sqrt(n);i+=2) if(n%i==0) return 0;
    return 1;
}
int main() {
    char name[]="madhiya zaid javed"; int roll=36; char uin[]="251M040"; char divc='F';
    int a,b,low,high;
    printf("%s Roll:%d UIN:%s Div:%c\n",name,roll,uin,divc);
    printf("Enter two integers (lower upper): ");
    if(scanf("%d %d",&a,&b)!=2) return 0;
    low = (a<b)?a:b; high = (a<b)?b:a;
    printf("Primes between %d and %d:\n", low, high);
    for(int i=low;i<=high;i++) if(isPrime(i)) printf("%d ", i);
    printf("\n");
    return 0;
}
```

Sample Input and Output (example run):

Input: 10 30

Output: Primes between 10 and 30:

11 13 17 19 23 29